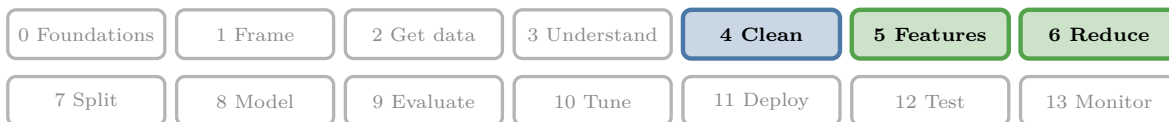


What is Feature Engineering

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Where this fits: Pipeline map steps 4 (Clean), 5 (Engineer features), 6 (Reduce dimensions). See the [Course map](#).



- **Builds on:** Dimensionality reduction ([Note 3](#)); Poor-quality data ([Note 9](#)); Features ([Note 11](#)); Train-test split ([Note 13](#)); Univariate analysis ([Note 20](#)); Correlation ([Note 21](#)).
- **Leads to:** Standardization ([Note 24](#)); Normalization ([Note 25](#)); Ordinal and label encoding ([Note 26](#)); Column transformer ([Note 28](#)); Function transformer ([Note 30](#)); Power transformer ([Note 31](#)).
- **Compare with:** Ordinal and label encoding ([Note 26](#)); Complete case analysis ([Note 35](#)); Random sample imputation ([Note 38](#)); KNN imputer ([Note 39](#)).

1. Overview

Key point: Feature engineering turns raw data into columns that a model can learn from well. It has four parts: transformation, construction, selection and extraction.

So far we can gather data and study it with EDA. Before a model can learn from that data, we have to prepare its columns. That work is feature engineering, and it fills the next stretch of Notes.

Figure 1 is the map of this whole stretch. Each of the four parts has its own techniques, and each technique gets its own Note later. This Note explains what each part is for, with one example each.

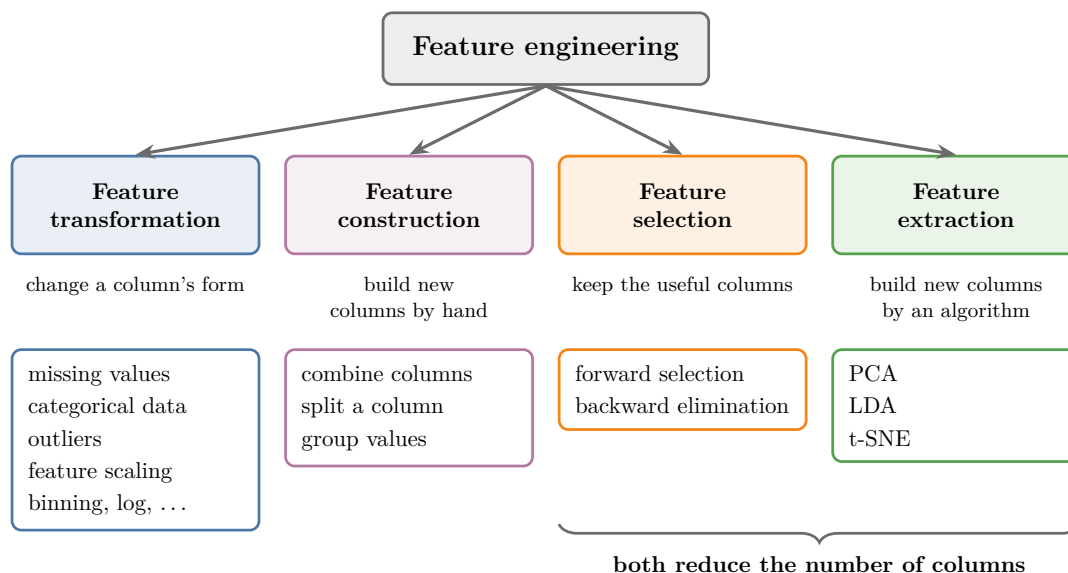


Figure 1: The four parts of feature engineering and their main techniques

2. What feature engineering is

Key point: Feature engineering uses knowledge of the problem to create, change and choose the columns given to an ML algorithm, so that it performs better.

Recall that a **feature** is one input column of the data. A common definition (from Wikipedia) reads:

Feature engineering is the process of using domain knowledge to extract features from raw data, which can be used to improve the performance of machine learning algorithms.

Two words in it matter:

- **Raw data:** the data as it arrives, with gaps, text, odd values and unhelpful columns. Having data does not mean we can hand it straight to an algorithm.
- **Domain knowledge:** knowledge of the field the data comes from, such as medicine, property or shipping. It tells us which columns make sense and which new ones would help.

A model gives good results only when its input columns are good. Turning raw data into such columns is feature engineering.

3. Why feature engineering matters

Key point: Good features with a simple algorithm usually beat poor features with a powerful algorithm.

A well-known saying in ML: give a weak algorithm excellent features and a powerful algorithm poor features, and the weak one usually wins. The columns we give a model limit what it can learn, however clever the model is.

Feature engineering is also more an art than a science. Like programming, it has known techniques, but each person combines them in their own way.

Two data scientists given the same data will often build different columns. This is why different books and courses teach it differently: there is no single fixed recipe. The techniques below are the shared toolkit, and experience decides how to use them.

4. Where feature engineering sits in an ML project

Key point: Feature engineering comes after the data is gathered, studied and given a first clean-up, and before any model is trained.

In the steps of an ML project, data first arrives and gets some initial preprocessing. Feature engineering comes next. Only then is the prepared data given to an algorithm to train a model.

Step	What happens
Get data	collect it from files, databases, APIs or the web
Understand data	study it with EDA
Engineer features	prepare the columns (this group of Notes)
Train a model	give the prepared data to an algorithm

Some lists of project steps name “feature engineering” and “feature selection” separately. Here both belong to feature engineering: selection is one of its four parts.

5. The four parts of feature engineering

Key point: We can change columns (transformation), add columns (construction), keep only some columns (selection), or replace columns with new ones made by an algorithm (extraction).

Figure 1 groups everything feature engineering does into four parts:

1. **Feature transformation:** change a column into another form, so that the model works better with it.
2. **Feature construction:** create a new column by hand, from existing ones, when we think it will give better results.
3. **Feature selection:** give the algorithm only the useful columns and drop the rest.
4. **Feature extraction:** let an algorithm build completely new columns out of the existing ones.

The next four sections take the parts in this order.

6. Feature transformation

Key point: Feature transformation changes a column’s form. Its four main jobs: fill missing values, turn categories into numbers, deal with outliers, and put columns on the same scale.

Sometimes a column is in a form that the algorithm cannot use, or cannot use well. Feature transformation changes it into a better form. It is the first step of feature engineering, and it has many techniques; these four are the most important.

6.1 Handling missing values

Key point: scikit-learn models do not accept missing values, so we either remove them or fill them in.

Real-world data almost always has gaps. A person may have left a form field blank, or something may have gone wrong while the data was collected.

Gaps are a real problem: scikit-learn, the main ML library in Python, refuses to train a model on data with missing values. So before training, we do one of two things:

- **Remove** the rows with gaps. This is fine when only a few rows are affected, since losing them hardly changes the data.
- **Fill** the gaps, when too many rows would be lost. For a numerical column we can use the column's mean or median; for a categorical column, its most common value (the **mode**).

Filling in missing values is called **imputation**. There are many more ways to do it, each with its own uses, and several Notes cover them one by one.

6.2 Handling categorical values

Key point: Algorithms work only with numbers, so text categories must be converted into numbers.

A categorical column holds labels rather than numbers, such as the names of animals (see the [types of ML Note](#), section 2.2). One common fix, one-hot encoding, replaces it with one 0/1 column per category; the [one-hot encoding Note](#) teaches it, and the [ordinal and label encoding Note](#) covers the other ways.

6.3 Binning: numbers into categories

Key point: Binning goes the other way: it groups a numerical column into ranges, which then act as categories.

Sometimes we convert numbers into categories instead. An **age** column holds exact ages, but the ranges may matter more than the exact values:

Age range	Group
0 to 15	child
16 to 30	young
31 to 45	adult
and so on	

Grouping numbers into ranges like this is called **binning**. It is one more way of changing a column's form.

6.4 Detecting and removing outliers

Key point: A few extreme values can pull a model far away from the pattern in the rest of the data.

An **outlier** is a value very different from the rest. In a class where most students score around 50 out of 100, one student scoring 98 is an outlier.

Outliers are risky because some algorithms are very sensitive to them. **Linear regression** draws the straight line that runs as close as possible to all the points. Figure 2 shows what three outliers do to it.

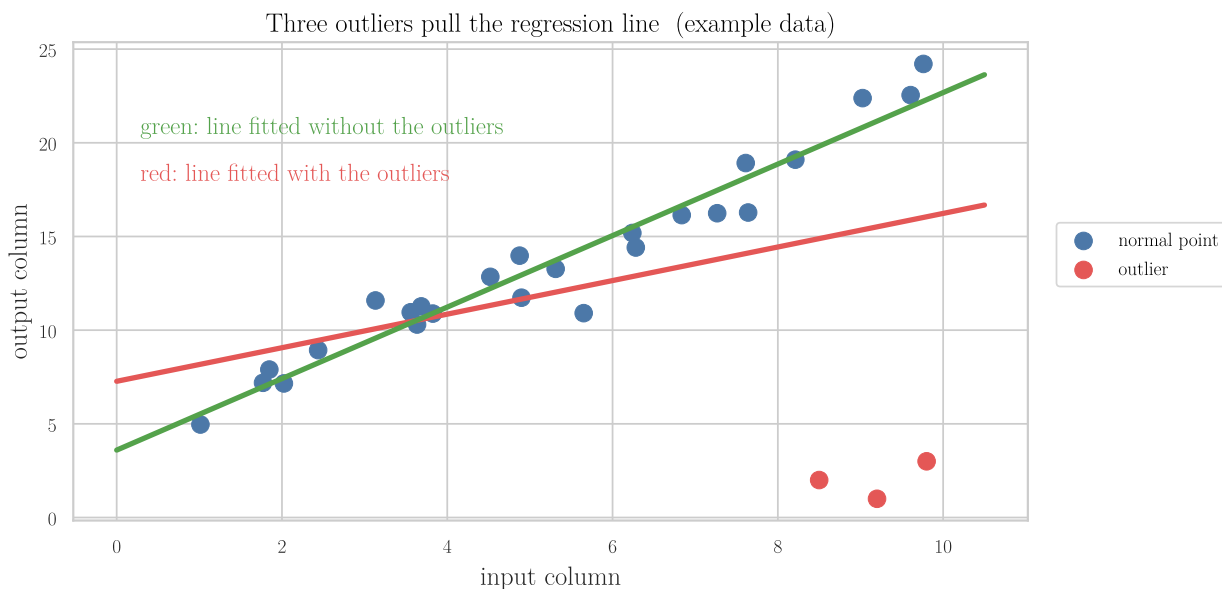


Figure 2: Three outliers pull the regression line away from the other points

- **Red line:** fitted to all points. To get close to the three outliers at the bottom right, it tilts away from the clear pattern of the other 25 points.
- **Green line:** fitted after removing the three outliers. It follows the main pattern.

When we use an algorithm that is sensitive to outliers, it is our job to deal with them before training. That means first **detecting** them, and then removing or adjusting them. Several Notes cover the methods.

6.5 Feature scaling

Key point: When columns have very different ranges, the column with the biggest numbers dominates; scaling puts all columns on a similar range.

When columns have very different ranges, such as **age** in the tens and **salary** in the tens of thousands, distance-based algorithms like KNN let the bigger column decide almost alone (see the scaling section of the [toy project Note](#), section 7). Feature scaling puts all columns on a similar range; its two main techniques, standardization and normalization, are taught in the [standardization Note](#) and the [normalization Note](#).

6.6 Other transformations

Key point: Many more transformations exist, such as the log transform and the Box-Cox transform.

Missing values, categories, outliers and scaling are the main jobs, but not the only ones. Mathematical transformations such as the **log transform** and the **Box-Cox transform** change the shape of a column's values. Dates and times, and columns that mix numbers with text, also need their own handling.

Extra: Missing values and outliers are also part of *cleaning* the data. On the Course map they sit at the Clean step, just before feature engineering. In practice the two steps overlap, and the same techniques serve both.

7. Feature construction

Key point: Feature construction creates a new column by hand from existing ones, guided by domain knowledge, intuition and experience.

Sometimes a useful column is not in the data at all. If we think such a column would give better results, we create it ourselves. This is feature construction.

There is no fixed method for it. What we build depends on how well we know the data, on intuition and on experience, so two people may build different columns.

7.1 Family size on the Titanic

Key point: Two Titanic columns, **SibSp** and **Parch**, combine into one clearer column: family size.

On the Titanic, **SibSp** (siblings and spouses on board) and **Parch** (parents and children on board) both describe the passenger's family, so we can add them into one **family_size** column and group it into alone, small and large families. The [feature construction Note](#), section 3, works this example through with code and measures whether it helps.

7.2 Splitting and grouping

Key point: Besides combining columns, we can also split one column into several, or group its values.

Feature construction has a few common patterns:

- **Combining:** several columns into one, as with family size.
- **Splitting:** one column into several, such as a full name into first name and title.
- **Grouping:** values into fewer categories, as with alone, small and large.

8. Feature selection

Key point: Feature selection keeps only the important columns and drops the rest, which makes the model better and faster.

Instead of giving the algorithm every column, we choose the useful ones and remove those that are not important. This is feature selection.

8.1 Pixels as columns: the MNIST dataset

Key point: In image data every pixel is a column, and many pixels, such as those at the edges, carry no information.

The **MNIST dataset** is a well-known collection of about 70,000 images of handwritten digits, 0 to 9. Each image is small: 28×28 pixels, so 784 pixels in all.

To use images in ML, each image is turned into one row of a table, with one column per pixel (Figure 3). The value in each column is that pixel's brightness. So the MNIST table has 784 input columns.

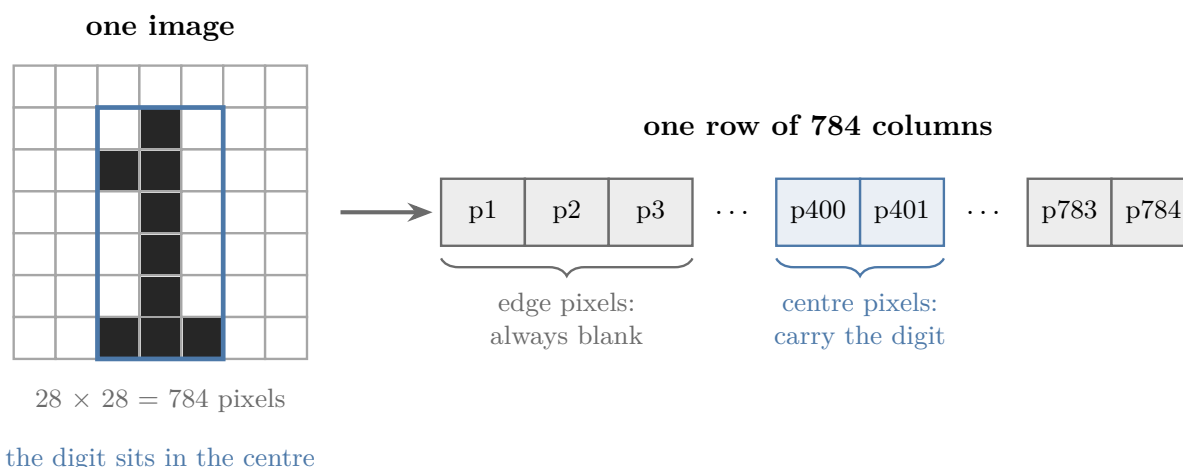


Figure 3: An image becomes one row with one column per pixel

Training on 784 columns takes a lot of time. And many of them are useless: the digit is always drawn in the centre, so the pixels near the edges are blank in nearly every image.

Feature selection keeps the important pixels, the ones in the centre, and drops the rest. The model then works with fewer columns, which brings two gains:

- **better performance**, since useless columns no longer get in the way;
- **more speed**, since there is less data to process.

Common techniques include **forward selection** (start with no columns, add the best one at a time) and **backward elimination** (start with all columns, remove the worst one at a time).

9. Feature extraction

Key point: Feature extraction builds completely new columns out of the existing ones, then keeps only the most useful new columns.

Feature extraction is the last part. Like feature construction, it creates new columns. Unlike construction, the new columns are computed by an algorithm, not chosen by hand.

The rooms and washrooms example of the [types of ML Note](#) (section 3.3, Dimensionality reduction) shows the idea: two related columns are replaced by one new column, the flat's area. An extraction algorithm does the same without domain knowledge: it builds the new columns from the data alone.

PCA (principal component analysis), the best-known extraction technique, rotates the axes so that a few new axes hold most of the information, and keeps those; the [PCA Note](#), section 4, shows the rotation step by step. If we start with 5 columns, PCA creates 5 new ones; we might keep the 2 most useful, so the number of columns goes down and none of the old columns is used directly.

The main extraction techniques are **PCA**, **LDA** (linear discriminant analysis) and **t-SNE**. They are especially useful for high-dimensional data, meaning data with very many columns.

10. The feature engineering Notes, in order

Key point: The Notes that follow work through the four parts in detail, starting with feature transformation.

Part	Topic	Notes
Transformation	Feature scaling: standardization, normalization	24, 25
Transformation	Encoding categorical data: ordinal, label, one-hot	26, 27
Transformation	Applying transformations to the right columns: column transformer, pipelines	28, 29
Transformation	Mathematical transforms: log, Box-Cox and others	30, 31
Transformation	Binning, mixed variables, dates and times	32 to 34
Transformation	Missing values: removing rows, imputation	35 to 40
Transformation	Outliers: what they are, detecting and removing them	41 to 44
Construction	Feature construction and splitting	45

Part	Topic	Notes
Extraction	Curse of dimensionality, PCA	46 to 49
Selection	Forward selection, backward elimination and others	later, after the main algorithms

The order of the Notes differs a little from the order of this Note. Each Note stands on its own, so it can also be read when the topic comes up in a project.

11. Summary

Part	What it does	Example	Main techniques
Feature transformation	changes a column's form	animal names into 0/1 columns	imputation, encoding, binning, outlier removal, scaling
Feature construction	builds a new column by hand	SibSp + Parch into family size	combining, splitting, grouping
Feature selection	keeps only the useful columns	drop the blank edge pixels of MNIST	forward selection, backward elimination
Feature extraction	builds new columns with an algorithm	rooms and washrooms into one PCA axis	PCA, LDA, t-SNE

- Feature engineering uses domain knowledge to turn raw data into columns that help a model perform better.
- Good features matter more than a powerful algorithm.
- It is partly an art: there are known techniques, but no single fixed recipe.
- It comes after the data is gathered and studied, and before a model is trained.
- Transformation's main jobs: missing values, categorical data, outliers and scaling.
- Selection and extraction both reduce the number of columns: selection keeps some old ones, extraction builds new ones.

12. Key terms

Term	Meaning
Feature engineering	Using domain knowledge to turn raw data into features that improve an ML model
Raw data	Data as it arrives, before any preparation
Domain knowledge	Knowledge of the field the data comes from
Feature transformation	Changing a column into a form the model can use better
Imputation	Filling in missing values, for example with the mean, median or mode
Mode	The most common value of a column
Binning	Grouping a numerical column into ranges that act as categories
Outlier	A value very different from the rest of the data
Linear regression	An algorithm that fits the straight line closest to all the points
Feature construction	Creating a new column by hand from existing ones

Term	Meaning
Feature selection	Keeping only the useful columns and dropping the rest
MNIST	A dataset of about 70,000 handwritten-digit images of 28×28 pixels
Feature extraction	Building completely new columns from the old ones with an algorithm
PCA	Principal component analysis: an extraction technique that rotates the axes and keeps the most useful new ones